

A sharp higher-order Cheeger inequality

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Abstract

Let $\lambda_k(G)$ be the k th eigenvalue of the normalized Laplacian of a finite undirected weighted graph G with positive degrees, where k is an integer satisfying $1 \leq k \leq |V(G)|$. Let $\varphi_k(G)$ be the minimum possible maximum conductance of k disjoint nonempty vertex sets. We prove

$$\varphi_k(G) \leq C \sqrt{\lambda_k(G) \log(k+1)}$$

for an absolute constant C . The number of sets and the spectral index are both k , and conductance is measured in the original graph. The logarithmic dependence is optimal up to an absolute constant. The proof combines geometric partitioning of the spectral embedding with minimum-cut improvement and adaptive projections in coefficient space. A single conductance threshold is used throughout the construction. The resulting maps have disjoint supports, and the sum of their Gram matrices is bounded below by an absolute positive multiple of the identity. A dyadic maximal estimate bounds the sum of their internal energies uniformly over unit coefficient vectors. A dimension argument using local eigenvalues then yields exactly k disjoint sparse cuts.

1 Introduction

Let $G = (V, w)$ be a finite undirected weighted graph, with symmetric nonnegative weights w_{uv} and positive degrees $d_v = \sum_u w_{uv}$. For disjoint sets $A, B \subseteq V$, write $w(A, B) = \sum_{u \in A, v \in B} w_{uv}$ and $w(v, B) = w(\{v\}, B)$. For $S \subseteq V$, write $S^c = V \setminus S$ and $\text{vol}(S) = \sum_{v \in S} d_v$. We also write $\text{cut}(S) = w(S, V \setminus S)$. For a nonempty set S , its edge conductance is

$$\varphi(S) = \frac{w(S, V \setminus S)}{\text{vol}(S)}.$$

We use this one-sided definition even when S has more than half the total volume. Define the k -way edge conductance by

$$\varphi_k(G) = \min_{\substack{\emptyset \neq S_1, \dots, S_k \subseteq V \\ S_i \cap S_j = \emptyset \ (i \neq j)}} \max_{1 \leq i \leq k} \varphi(S_i). \quad (1.1)$$

The sets need not cover V . Let $0 = \lambda_1(G) \leq \dots \leq \lambda_{|V|}(G)$ be the eigenvalues of the normalized Laplacian. We omit G from $\lambda_j(G)$ when the graph is clear. For a nonzero real function f on V , its Rayleigh quotient is

$$\frac{\sum_{\{u,v\}} w_{uv} (f(u) - f(v))^2}{\sum_v d_v f(v)^2},$$

where the numerator counts each unordered nonloop edge once.

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Lee, Oveis Gharan, and Trevisan [3] proved

$$\frac{\lambda_k(G)}{2} \leq \varphi_k(G) \leq O(k^2) \sqrt{\lambda_k(G)}.$$

The dependence on k improves when fewer sets are required. Louis, Raghavendra, Tetali, and Vempala [5] and Lee, Oveis Gharan, and Trevisan [3] obtain a constant fraction of k disjoint sets with conductance $O(\sqrt{\lambda_k \log k})$. In particular, the latter work gives $\varphi_k(G) \leq O(\sqrt{\lambda_{2k} \log k})$ when $2k \leq |V|$. The question is whether the same logarithmic dependence holds when the number of sets and the eigenvalue index are both k .

Makarychev, Makarychev, Shan, and Vijayaraghavan [6] prove related inequalities for partitions with buffers, where edges to small buffer sets are excluded from the boundary. Their logarithmic estimate also allows a larger eigenvalue index.

Lau's book manuscript [2, Theorem 7.17] presents the bound

$$\varphi_k(G) \leq O((\log k)^{3/2}) \sqrt{\lambda_k(G)}. \quad (1.2)$$

Its proof repeatedly improves the current family of sparse cuts. Each round reduces the deficit in the number of sets by a constant factor and adds $O(\sqrt{\lambda_k \log k})$ to the conductance bound. After $O(\log k)$ rounds, the family contains k sets. In an earlier paper [4], the author obtained a weaker polylogarithmic bound through localization of the spectral embedding and adaptive coefficient projections. That method also gives disjoint functions with small original-graph Rayleigh quotients.

Here we combine cut improvement with adaptive spectral localization to improve the bound in (1.2) by a factor of $\log k$. The number of sets and the eigenvalue index remain equal to k .

Theorem 1.1. *There is an absolute constant C such that, for every finite undirected weighted graph $G = (V, w)$ with positive degrees and every integer $1 \leq k \leq |V|$, there exist k pairwise disjoint nonempty sets $S_1, \dots, S_k$ satisfying*

$$\max_{1 \leq i \leq k} \varphi(S_i) \leq C \sqrt{\lambda_k(G) \log(k+1)}. \quad (1.3)$$

The exponent $1/2$ on the logarithm cannot be decreased. The noisy hypercube example in [3, Section 4.4] gives graphs for which $\varphi_k(G)/\sqrt{\lambda_k(G)} \geq c\sqrt{\log k}$. We verify the lower bound with spectral index k in Section 7.

1.1 Proof overview

Let $F : \mathbb{R}^k \rightarrow L^2(d)$ have as columns eigenfunctions for the first k eigenvalues of the random-walk Laplacian, orthonormal in the degree-weighted inner product. Every nonzero function in its range has Rayleigh quotient at most λ_k .

A fixed conductance threshold. We use one threshold $\tau = C_0 \sqrt{\lambda_k \log(k+1)}$ throughout, where C_0 is a sufficiently large absolute constant. For a vertex set B , choose $S \subseteq B$ to minimize

$$\text{cut}(T) - \tau \text{vol}(T), \quad T \subseteq B.$$

If S is nonempty, then $\varphi(S) \leq \tau$. Comparison with $S \setminus T$ also shows that every $T \subseteq S$ satisfies

$$w(T, S^c) \leq \tau \text{vol}(T) + w(T, S \setminus T). \quad (1.4)$$

This is the boundary inequality used in Lau's local-to-global lemma for cores [2, Lemma 7.18], with τ in place of $\varphi(S)$. For nonempty T , it implies $\varphi(T) \leq \tau + 2\varphi_S(T)$, where $\varphi_S(T)$ is local conductance with original degrees in the denominator. The same minimization also controls how much spectral mass is rejected.

Keeping the remaining directions. At each stage, we work in a subspace of coefficient space. Geometric partitioning and cut improvement select disjoint sets from the unassigned vertices. Vertices that are not selected remain available. We then retain the spectral subspace of the remaining Gram matrix corresponding to eigenvalues close to one. The corresponding spectral functions have little mass on previously selected sets. The inequality (1.4) then controls the boundary contributions from those sets without increasing τ .

A trace estimate suffices to reduce the dimension of this subspace by a constant factor. Because selection is by vertex sets, the Gram matrices on the selected and remaining vertices add exactly. The discarded coefficient spaces are orthogonal, so after $O(\log k)$ stages the sum of the Gram matrices of the selected maps is bounded below by an absolute positive multiple of the identity. Each individual Gram matrix has a small sum of eigenvalues other than the largest.

Obtaining exactly k sets. The selected maps use different, nested coefficient projections. A dyadic maximal estimate bounds the sum of their internal energies, uniformly over unit coefficient vectors, by $O((1 + \log \log(k + 2))^2 \lambda_k)$. A dimension argument then shows that the local eigenvalue counts below a constant multiple of this bound, each truncated at two, sum to at least k . Local Cheeger rounding gives one or two sets from each selected vertex set. The boundary inequality converts their local conductance to original conductance at cost

$$\tau + O((1 + \log \log(k + 2))\sqrt{\lambda_k}),$$

which is $O(\sqrt{\lambda_k \log(k + 1)})$.

The proof follows this order. Section 2 fixes the local operators and spectral notation. Sections 3 and 4 establish cut improvement and geometric partitioning. Section 5 constructs the selected maps, and Section 6 completes the proof.

2 Notation and preliminaries

Self-loops are allowed and count once in degree. Sums indexed by $\{u, v\}$ count each unordered nonloop edge once. Let $A = (w_{uv})$, $D = \text{diag}(d_v)$, and

$$\mathcal{L} = D^{-1/2}(D - A)D^{-1/2}.$$

We write $L^2(d)$ for the space of real functions on V with inner product $\langle f, g \rangle_d = \sum_v d_v f(v)g(v)$, induced norm $\|f\|_d^2 = \langle f, f \rangle_d$, and energy

$$\mathcal{E}(f) = \sum_{\{u, v\}} w_{uv} (f(u) - f(v))^2.$$

The random-walk Laplacian $\Delta = D^{-1}(D - A)$ is self-adjoint in this inner product and similar to $\mathcal{L}$. In this notation, the Rayleigh quotient of a nonzero function f is $\mathcal{E}(f)/\|f\|_d^2$. For vector-valued maps, energy is the sum of coordinate energies. All logarithms are natural except $\log_2$. Constants denoted by C are positive and absolute unless a dependence is indicated; their values may increase between estimates.

Local conductance and local eigenvalues. For a nonempty $S \subseteq V$ and nonempty $T \subseteq S$, define

$$\varphi_S(T) = \frac{w(T, S \setminus T)}{\text{vol}(T)}, \quad \mathcal{E}_S(f) = \sum_{\{u, v\} \subset S} w_{uv} (f(u) - f(v))^2.$$

The local graph keeps the internal edges of S and adds the outgoing weight $w(v, S^c)$ to the loop at v . Thus its degrees are the original d_v . Its eigenvalues, denoted by $\lambda_j(S)$, are those of

$$(\Delta_S f)(v) = \frac{1}{d_v} \sum_{u \in S} w_{vu} (f(v) - f(u))$$

in $L^2(d|_S)$. Norms and inner products of functions on S use the restricted degree weights $d|_S$. In particular, $\lambda_1(S) = 0$. This is the weighted version of the local operator used in [2, Section 7.7]; it is different from the Dirichlet operator, which includes boundary energy.

Spectral embedding and Gram matrices. Let $F : \mathbb{R}^k \rightarrow L^2(d)$ have as columns eigenfunctions for the first k eigenvalues of Δ , orthonormal in the degree-weighted inner product. Adjoints are taken with respect to the specified inner products; in particular, $F^* = F^T D$. Let $\mathbf{f}_v = F^T e_v \in \mathbb{R}^k$ be the transpose of the row of F indexed by v , where e_v is the corresponding coordinate vector. The map $v \mapsto \mathbf{f}_v$ is the spectral embedding, while $F : \mathbb{R}^k \rightarrow L^2(d)$ acts on coefficient vectors by $(Fx)(v) = \langle \mathbf{f}_v, x \rangle$. Then

$$F^* F = I_k, \quad \mathcal{E}(Fx) \leq \lambda_k \|x\|^2. \quad (2.1)$$

For a matrix $Z \in \mathbb{R}^{k \times \ell}$ with orthonormal columns, $Z^T Z = I_\ell$, put $\mathbf{y}_v = Z^T \mathbf{f}_v$. The columns of Z form a basis of the active coefficient space, and FZ represents F on that space. Then

$$\sum_v d_v \mathbf{y}_v \mathbf{y}_v^T = I_\ell, \quad \mathcal{E}(FZ) = \sum_{\{u,v\}} w_{uv} \|\mathbf{y}_u - \mathbf{y}_v\|^2 \leq \ell \lambda_k. \quad (2.2)$$

For a set B , let $\mathbf{1}_B$ denote both its indicator function and the operator of multiplication by that function. The matrix $Z^T F^* \mathbf{1}_B F Z = \sum_{v \in B} d_v \mathbf{y}_v \mathbf{y}_v^T$ is the weighted Gram matrix of the columns of FZ restricted to B . Matrix norms are operator norms and vector norms are Euclidean unless indicated otherwise. For symmetric matrices, $M \preceq N$ means that $N - M$ is positive semidefinite.

For a positive semidefinite matrix M , we use $\text{tail}_1(M)$ as shorthand for the sum of its eigenvalues other than the largest. Thus

$$\text{tail}_1(M) = \text{tr } M - \lambda_{\max}(M) = \min_{\|u\|=1} \text{tr}((I - uu^T)M). \quad (2.3)$$

The variational formula shows that $0 \preceq M' \preceq M$ implies $\text{tail}_1(M') \leq \text{tail}_1(M)$.

Coarea and threshold rounding. For a real function h , define its total variation by $\text{TV}(h) = \sum_{\{u,v\}} w_{uv} |h(u) - h(v)|$. For $h \geq 0$, coarea gives

$$\text{TV}(h) = \int_0^\infty \text{cut}(\{h > t\}) dt, \quad \sum_v d_v h(v) = \int_0^\infty \text{vol}(\{h > t\}) dt. \quad (2.4)$$

For every real f , Cauchy–Schwarz gives

$$\text{TV}(f^2) \leq \sqrt{\mathcal{E}(f) \sum_{\{u,v\}} w_{uv} (f(u) + f(v))^2} \leq \sqrt{2\mathcal{E}(f) \|f\|_d^2}. \quad (2.5)$$

Write $f_+ = \max\{f, 0\}$ and $f_- = \max\{-f, 0\}$. A nonzero nonnegative function f has a nonempty superlevel set $T \subseteq \text{supp } f$ with $\varphi(T) \leq \sqrt{2\mathcal{E}(f)}/\|f\|_d$. The same statement holds for the local graph on S , with $\mathcal{E}_S$ and φ_S . This is the Cheeger rounding step in [2, Lemma 7.4].

3 Cut improvement at a fixed threshold

We isolate the boundary property needed for local-to-global rounding.

Definition 3.1. For $\tau \geq 0$, a nonempty set S is τ -stable if

$$w(T, S^c) \leq \tau \operatorname{vol}(T) + w(T, S \setminus T) \quad \text{for every } T \subseteq S. \quad (3.1)$$

Taking $T = S$ shows that $\varphi(S) \leq \tau$. More generally,

$$\varphi(T) \leq \tau + 2\varphi_S(T) \quad (\emptyset \neq T \subseteq S). \quad (3.2)$$

A core of conductance at most τ , in the terminology of [2, Section 7.7], is τ -stable by the proof of Lemma 7.18 there. We need only (3.1), rather than the full minimum-conductance property of a core.

Lemma 3.2 (Cut improvement). *Fix $B \subseteq V$ and $\tau > 0$. Let $S \subseteq B$, with the empty set allowed, minimize*

$$J(T) = \operatorname{cut}(T) - \tau \operatorname{vol}(T), \quad T \subseteq B. \quad (3.3)$$

If S is nonempty, then it is τ -stable. For every real function g supported in B ,

$$\tau \|\mathbf{1}_{B \setminus S} g\|_d^2 + \operatorname{TV}((\mathbf{1}_S g)^2) \leq \operatorname{TV}(g^2). \quad (3.4)$$

Proof. For $T \subseteq S$, comparison with $S \setminus T$ gives

$$0 \leq J(S \setminus T) - J(S) = w(T, S \setminus T) - w(T, S^c) + \tau \operatorname{vol}(T),$$

which is (3.1). For arbitrary $T \subseteq B$, comparison with $S \cup T$ and submodularity of the cut function give

$$\tau \operatorname{vol}(T \setminus S) \leq \operatorname{cut}(S \cup T) - \operatorname{cut}(S) \leq \operatorname{cut}(T) - \operatorname{cut}(S \cap T).$$

Apply this to the superlevel sets of g^2 and integrate using (2.4). Their intersections with S are the superlevel sets of $(\mathbf{1}_S g)^2$, proving (3.4). $\square$

The minimization in (3.3) is a source–sink minimum cut, as in [2, Problem 7.24]. For each $v \in B$, add a source arc of capacity τd_v and a sink arc of capacity $w(v, B^c)$, and keep the internal undirected edges with their original capacities. The cut with source-side vertices $T \subseteq B$ has capacity $\tau \operatorname{vol}(B) + J(T)$.

Lemma 3.3 (Boundary mass). *If S is τ -stable, then every function g on S satisfies*

$$\sum_{v \in S} w(v, S^c) g(v)^2 \leq \tau \|g\|_d^2 + \sqrt{2\mathcal{E}_S(g)} \|g\|_d^2. \quad (3.5)$$

Proof. Integrating (3.1) over the superlevel sets of g^2 gives

$$\sum_{v \in S} w(v, S^c) g(v)^2 \leq \tau \|g\|_d^2 + \sum_{\{u,v\} \subset S} w_{uv} |g(u)^2 - g(v)^2|.$$

Apply (2.5) to the internal edges. At each vertex, the total weight of incident internal edges is at most its original degree, so the same norm applies. $\square$

4 Geometric partitioning of the embedding

This section constructs a random partition of the embedding and bounds the expected total mass of cells whose Gram matrices have a large sum of eigenvalues other than the largest. We also bound the weighted cost of edges separated by the partition. The geometric ingredient is the Euclidean random-ball partition of Charikar, Chekuri, Goel, Guha, and Plotkin [1, Section 3.1]. We include the needed argument.

Lemma 4.1 (Random balls). *For every integer $q \geq 1$, every finite set of points in $\mathbb{R}^q$, and every $r > 0$, there is a random partition into cells of diameter at most $2r$ such that*

$$\mathbb{P}[B(x) \neq B(y)] \leq \min\{1, 2\sqrt{q}\|x - y\|/r\}, \quad (4.1)$$

where $B(x)$ denotes the cell containing x .

Proof. The empty point set is trivial, so assume the point set is nonempty. Sample independent centers uniformly from the union of the radius- r balls centered at the given points, and assign each point to the first sampled ball that contains it. All points are covered almost surely, and each cell is contained in one radius- r ball. The centers are sampled from the whole union of balls, not from the finite set of given points.

Write $\mathbb{B}(x, r)$ for the Euclidean ball of radius r centered at x . For two points x, y , the first sampled center in $\mathbb{B}(x, r) \cup \mathbb{B}(y, r)$ is uniform in that union. The points lie in different cells exactly when this center belongs to the symmetric difference. Hence

$$\mathbb{P}[B(x) \neq B(y)] = \frac{|\mathbb{B}(x, r) \triangle \mathbb{B}(y, r)|}{|\mathbb{B}(x, r) \cup \mathbb{B}(y, r)|}.$$

Slicing parallel to $x - y$ bounds the numerator by $2\|x - y\|v_{q-1}r^{q-1}$, where v_q is the volume of the unit q -ball. The denominator is at least $v_q r^q$. Finally, $v_{q-1}/v_q \leq \sqrt{q}$. For $q \geq 2$, this follows from

$$\frac{v_q}{v_{q-1}} = \int_{-1}^1 (1 - t^2)^{(q-1)/2} dt \geq \frac{1}{\sqrt{q}};$$

restrict to $|t| \leq 1/\sqrt{q}$ and use $(1 - 1/q)^{(q-1)/2} \geq 1/2$. The case $q = 1$ is immediate. $\square$

Lemma 4.2 (Partitioning the spectral embedding). *Fix $0 < \delta, \beta < 1$. Let $Z : \mathbb{R}^\ell \rightarrow \mathbb{R}^k$ be an isometry with $1 \leq \ell \leq k$, and let $R \subseteq V$. Use the vectors $\mathbf{y}_v = Z^T \mathbf{f}_v$ from (2.2). There is a random partition of $\{v \in R : \mathbf{y}_v \neq 0\}$ into cells B . Write $B(v)$ for the cell containing v and $\Sigma_B = \sum_{v \in B} d_v \mathbf{y}_v \mathbf{y}_v^T$. Then*

$$\mathbb{E} \sum_{B: \text{tail}_1(\Sigma_B) > \delta} \text{tr } \Sigma_B \leq \beta \ell, \quad (4.2)$$

and

$$\mathbb{E} \sum_{\substack{\{u, v\} \subset R: \mathbf{y}_u, \mathbf{y}_v \neq 0 \\ B(u) \neq B(v)}} w_{uv} (\|\mathbf{y}_u\|^2 + \|\mathbf{y}_v\|^2) \leq C_{\delta, \beta} \ell \sqrt{\lambda_k \log(k+1)}. \quad (4.3)$$

Proof. Put $m_v = d_v \|\mathbf{y}_v\|^2$ for every vertex and $a_v = \mathbf{y}_v / \|\mathbf{y}_v\|$ for nonzero rows. Sums involving a_v omit zero rows. Global orthonormality gives

$$\sum_v m_v a_v a_v^T = I_\ell, \quad \sum_v m_v = \ell. \quad (4.4)$$

Set $\eta = \sqrt{\delta/[2(1+\delta)]}$, put $r = \eta/20$, and choose

$$q = \lceil \max\{1, \log(2k/(\delta\beta))\} \rceil.$$

Let $\Gamma : \mathbb{R}^\ell \rightarrow \mathbb{R}^q$ have independent $\mathcal{N}(0, 1/q)$ entries. Apply Lemma 4.1 to the unnormalized projected points Γa_v for $v \in R$ with $\mathbf{y}_v \neq 0$. For fixed δ, β , this choice gives $q = O_{\delta, \beta}(\log(k+1))$.

Eigenvalues of the cell Gram matrices. If $\|a_u - a_v\| > \eta$ and u, v belong to one cell, then $\|\Gamma(a_u - a_v)\| \leq 2r < \|a_u - a_v\|/10$. For fixed $x \neq 0$, the random variable $q\|\Gamma x\|^2/\|x\|^2$ has the χ_q^2 distribution. The lower-tail Chernoff bound gives

$$\mathbb{P}[\|\Gamma x\| \leq \|x\|/10] \leq e^{-q}. \quad (4.5)$$

For completeness, writing χ_q^2 for a random variable with this distribution, Markov's inequality gives

$$\mathbb{P}[\chi_q^2 \leq q/100] \leq e^{0.495q} \mathbb{E} e^{-49.5\chi_q^2} = e^{0.495q} 100^{-q/2} \leq e^{-q}.$$

For a cell B , define the total weight of ordered pairs at distance greater than η by

$$W_B = \sum_{\substack{u, v \in B \\ \|a_u - a_v\| > \eta}} m_u m_v.$$

It follows from (4.5) that

$$\mathbb{E} \sum_B W_B \leq e^{-q} \ell^2. \quad (4.6)$$

Call a cell B bad if $\text{tail}_1(\Sigma_B) > \delta$, and good otherwise. Write $m_B = \text{tr } \Sigma_B = \sum_{v \in B} m_v$. Since $\Sigma_B \preceq I_\ell$ by (4.4), we have $m_B \leq 1 + \text{tail}_1(\Sigma_B)$. Also,

$$m_B \text{tail}_1(\Sigma_B) \leq m_B^2 - \text{tr}(\Sigma_B^2) = \sum_{u, v \in B} m_u m_v (1 - \langle a_u, a_v \rangle^2) \leq \eta^2 m_B^2 + W_B.$$

The first inequality uses $\text{tr}(\Sigma_B^2) \leq \lambda_{\max}(\Sigma_B) \text{tr } \Sigma_B$. For the last inequality, use $1 - \langle a_u, a_v \rangle^2 \leq \|a_u - a_v\|^2$ for pairs at distance at most η , and $1 - \langle a_u, a_v \rangle^2 \leq 1$ for the remaining pairs. If $\text{tail}_1(\Sigma_B) > \delta$, then $\eta^2 m_B \leq \eta^2(1 + \text{tail}_1(\Sigma_B)) \leq \text{tail}_1(\Sigma_B)/2$. Thus $W_B \geq \delta m_B/2$ for every bad cell. Summing and using (4.6) proves

$$\mathbb{E} \sum_{B: \text{tail}_1(\Sigma_B) > \delta} \text{tr } \Sigma_B \leq \frac{2}{\delta} e^{-q} \ell^2 \leq \beta \ell.$$

Boundary cost. Conditional on Γ , Lemma 4.1 gives

$$\mathbb{P}[B(u) \neq B(v) \mid \Gamma] \leq \frac{2\sqrt{q}}{r} \|\Gamma(a_u - a_v)\|.$$

Since $\mathbb{E}\|\Gamma x\| \leq \|x\|$, averaging over Γ bounds the separation probability by $(2\sqrt{q}/r)\|a_u - a_v\|$. For unit vectors a, b and $s, t \geq 0$,

$$(s^2 + t^2)\|a - b\| \leq 2(s + t)\|sa - tb\|. \quad (4.7)$$

Indeed, the triangle inequality gives

$$(s+t)\|a-b\| \leq \|sa-tb\| + \|ta-sb\| = 2\|sa-tb\|,$$

where the equality follows from $\|a\| = \|b\| = 1$. Multiplying by $s+t$ and using $s^2 + t^2 \leq (s+t)^2$ proves (4.7). Therefore the expectation in (4.3) is at most

$$\frac{4\sqrt{q}}{r} \sum_{\{u,v\}} w_{uv}(\|\mathbf{y}_u\| + \|\mathbf{y}_v\|)\|\mathbf{y}_u - \mathbf{y}_v\| \leq \frac{4\sqrt{q}}{r} \sqrt{2\ell \mathcal{E}(FZ)} \leq \frac{4\sqrt{2q}}{r} \ell \sqrt{\lambda_k}.$$

Here Cauchy–Schwarz uses $\sum_{\{u,v\}} w_{uv}(\|\mathbf{y}_u\| + \|\mathbf{y}_v\|)^2 \leq 2\ell$. This proves the desired bound. $\square$

Discarding whole cells with $\text{tail}_1(\Sigma_B) > \delta$ creates no new boundary inside a retained cell. The vertices of the discarded cells remain available for later stages. We do not discard individual vertices according to the norm of their projected vectors.

5 Adaptive selection of vertex sets

For the construction, assume $k \geq 2$ and $\lambda_k > 0$. The remaining cases are handled in the proof of Theorem 1.1. Throughout the construction, G and k are fixed. Set

$$\varepsilon = \frac{1}{64}, \quad \delta = \frac{\varepsilon}{4}, \quad \beta = \varepsilon, \quad \tau = C_0 \sqrt{\lambda_k \log(k+1)}, \quad (5.1)$$

where the absolute constant C_0 is chosen in the proof of Lemma 5.1 below. The threshold τ depends on G and k but remains fixed throughout the construction.

5.1 One selection step

Lemma 5.1. *Suppose previously selected sets are pairwise disjoint and τ -stable, and let R be the complement of their union. Let $Z : \mathbb{R}^\ell \rightarrow \mathbb{R}^k$ be an isometry, where $1 \leq \ell \leq k$, such that*

$$Z^T F^* \mathbf{1}_R FZ \succeq (1 - \varepsilon) I_\ell. \quad (5.2)$$

One can select additional disjoint nonempty sets $S_i \subseteq R$, each τ -stable and satisfying

$$\text{tail}_1(Z^T F^* \mathbf{1}_{S_i} FZ) \leq \delta,$$

so that, for $R' = R \setminus \bigcup_i S_i$,

$$\text{tr}(Z^T F^* \mathbf{1}_{R'} FZ) \leq \ell/4. \quad (5.3)$$

Proof. Put $\alpha(v) = \|\mathbf{y}_v\|$, where $\mathbf{y}_v = Z^T \mathbf{f}_v$. Then $\|\alpha\|_d^2 = \ell$. The reverse triangle inequality gives $|\alpha(u) - \alpha(v)| \leq \|\mathbf{y}_u - \mathbf{y}_v\|$ on every edge, and hence $\mathcal{E}(\alpha) \leq \mathcal{E}(FZ) \leq \ell \lambda_k$. Equation (5.2) implies

$$\sum_{v \notin R} d_v \alpha(v)^2 \leq \varepsilon \ell. \quad (5.4)$$

Apply Lemma 3.3 to α on each previously selected set S . Their internal edge sets are disjoint, so Cauchy–Schwarz gives

$$\sum_S \sqrt{2\mathcal{E}_S(\alpha) \|\mathbf{1}_S \alpha\|_d^2} \leq \sqrt{2 \left(\sum_S \mathcal{E}_S(\alpha) \right) \left(\sum_S \|\mathbf{1}_S \alpha\|_d^2 \right)} \leq \ell \sqrt{2\varepsilon \lambda_k},$$

where the sums run over the previously selected sets, and the last inequality uses (5.4) and $\mathcal{E}(\alpha) \leq \ell \lambda_k$. Summing (3.5) therefore gives

$$\sum_{u \notin R, v \in R} w_{uv} \alpha(u)^2 \leq \tau \varepsilon \ell + \ell \sqrt{2\varepsilon \lambda_k}. \quad (5.5)$$

The sum on the left of (5.5) is no larger than the sum of the left sides of (3.5) over the previously selected sets; edges between two such sets may also appear in the latter sum.

For an edge from $v \in R$ to $u \notin R$, the inequality $\alpha(v)^2 \leq |\alpha(v)^2 - \alpha(u)^2| + \alpha(u)^2$ shows that

$$\text{TV}(\mathbf{1}_R \alpha^2) \leq \text{TV}(\alpha^2) + \sum_{u \notin R, v \in R} w_{uv} \alpha(u)^2. \quad (5.6)$$

By (2.5), $\text{TV}(\alpha^2) \leq \ell \sqrt{2\lambda_k}$.

Apply Lemma 4.2 to the nonzero rows in R . Placing the endpoints of an edge inside R in different cells increases its contribution to the sum of variations by at most $w_{uv}(\alpha(u)^2 + \alpha(v)^2)$. If either row is zero, there is no increase. Hence

$$\mathbb{E} \sum_B \text{TV}(\mathbf{1}_B \alpha^2) \leq \ell \left(\tau \varepsilon + \sqrt{2\varepsilon \lambda_k} + \sqrt{2\lambda_k} + C_{\delta, \beta} \sqrt{\lambda_k \log(k+1)} \right). \quad (5.7)$$

Discard whole cells whose Gram matrices satisfy $\text{tail}_1(\Sigma_B) > \delta$. Their expected mass is at most $\beta \ell$. In each remaining cell B , apply Lemma 3.2 and let S_B be its selected subset. Keep the nonempty sets S_B . Each is τ -stable, and its Gram matrix satisfies $\text{tail}_1(Z^T F^* \mathbf{1}_{S_B} F Z) \leq \delta$ by (2.3).

The vertices remaining in R' are those with zero rows, those in bad cells, and those rejected inside good cells. Applying (3.4) with $g = \mathbf{1}_B \alpha$ bounds the mass of the last group. Thus, using (5.7) and (5.1),

$$\mathbb{E} \text{tr}(Z^T F^* \mathbf{1}_{R'} F Z) \leq \beta \ell + \frac{1}{\tau} \mathbb{E} \sum_{B \text{ good}} \text{TV}(\mathbf{1}_B \alpha^2) \leq \ell \left(\beta + \varepsilon + \frac{C_{\delta, \beta}}{C_0} + \frac{\sqrt{2\varepsilon} + \sqrt{2}}{C_0 \sqrt{\log(k+1)}} \right).$$

Since $\beta + \varepsilon = 1/32$ and $\log(k+1) \geq \log 3$, it suffices to choose

$$C_0 \geq \frac{32}{7} \left(C_{\delta, \beta} + \frac{\sqrt{2\varepsilon} + \sqrt{2}}{\sqrt{\log 3}} \right).$$

Here $C_{\delta, \beta}$ is the constant in (4.3); the fixed values of $\delta, \beta, \varepsilon$ make this choice independent of the stage and of k . With this choice, the expectation above is at most $(1/32 + 7/32)\ell = \ell/4$. Some outcome therefore satisfies (5.3). All other assertions hold for the same outcome. No independence between successive selection steps is needed. $\square$

5.2 The sum of the Gram matrices

Start with $R_0 = V$ and the coefficient projector $\Pi_0 = I_k$. At step j , choose an orthonormal basis of $\text{ran } \Pi_{j-1}$ and apply Lemma 5.1 to R_{j-1} using the corresponding isometry. Let R_j be the vertices of R_{j-1} left after removing the newly selected sets. Define the Gram matrices before and after this selection, in the same coefficient space, by

$$K_j = \Pi_{j-1} F^* \mathbf{1}_{R_{j-1}} F \Pi_{j-1}, \quad Q_j = \Pi_{j-1} F^* \mathbf{1}_{R_j} F \Pi_{j-1}.$$

The newly selected sets partition $R_{j-1} \setminus R_j$, and

$$K_j - Q_j = \Pi_{j-1} F^* \mathbf{1}_{R_{j-1} \setminus R_j} F \Pi_{j-1} \succeq 0. \quad (5.8)$$

Let Π_j be the spectral projector of Q_j , within $\text{ran } \Pi_{j-1}$, for eigenvalues strictly greater than $1 - \varepsilon$. Then $\Pi_j \preceq \Pi_{j-1}$, and any isometry onto $\text{ran } \Pi_j$ satisfies (5.2) with $R = R_j$. If $\ell_j = \text{rank } \Pi_j$, then

$$\ell_j \leq \frac{\text{tr } Q_j}{1 - \varepsilon} \leq \frac{\ell_{j-1}}{4(1 - \varepsilon)}. \quad (5.9)$$

Let N be the first step for which $\Pi_N = 0$. Iterating (5.9) gives

$$\ell_j \leq k[4(1 - \varepsilon)]^{-j}, \quad N \leq 1 + \left\lfloor \frac{\log k}{\log(4(1 - \varepsilon))} \right\rfloor = O(\log(k + 1)),$$

since a nonnegative integer rank smaller than one is zero.

For each selected set S_i produced at step j , define the localized map $F_i : \mathbb{R}^k \rightarrow L^2(d)$ by

$$F_i x = \mathbf{1}_{S_i} F \Pi_{j-1} x. \quad (5.10)$$

The maps have pairwise disjoint supports. Their weighted Gram matrices satisfy $\text{tail}_1(F_i^* F_i) \leq \delta$: passing from the current coefficient space to $\mathbb{R}^k$ adds only zero eigenvalues to the Gram matrix of a selected set. By (5.8), summing $F_i^* F_i$ over the sets selected at step j gives $K_j - Q_j$.

Lemma 5.2. *The localized maps satisfy*

$$\varepsilon I_k \preceq \sum_i F_i^* F_i \preceq I_k. \quad (5.11)$$

Proof. Put $P_j = \Pi_{j-1} - \Pi_j$ and set $K_{N+1} = 0$. Since the projectors are nested, $\Pi_j \Pi_{j-1} = \Pi_j$. Also, Π_j commutes with Q_j because it is a spectral projector of Q_j . Thus

$$\Pi_j Q_j \Pi_j = \Pi_j F^* \mathbf{1}_{R_j} F \Pi_j = K_{j+1}, \quad \Pi_j Q_j P_j = Q_j \Pi_j P_j = 0.$$

By symmetry the other off-diagonal block also vanishes. The eigenvalues of Q_j on $\text{ran } P_j$ are at most $1 - \varepsilon$, so

$$Q_j = K_{j+1} + P_j Q_j P_j, \quad 0 \preceq P_j Q_j P_j \preceq (1 - \varepsilon) P_j.$$

Since $K_1 = I_k$, summing this identity gives

$$\sum_i F_i^* F_i = \sum_{j=1}^N (K_j - Q_j) = I_k - \sum_{j=1}^N P_j Q_j P_j.$$

The projectors P_j have mutually orthogonal ranges and sum to I_k . Consequently $0 \preceq \sum_j P_j Q_j P_j \preceq (1 - \varepsilon) I_k$, which proves (5.11). $\square$

The only commutation used above is that of Q_j with its own spectral projector; Gram matrices of different cells need not commute.

6 Internal energy and exactly k sets

The internal edges of the selected sets are disjoint. This allows a single maximal estimate to control all coefficient projections.

Lemma 6.1 (A dyadic maximal estimate). *Let $I_k = \Pi_0 \succeq \cdots \succeq \Pi_N = 0$ be nested orthogonal projectors, and let $\mathcal{T} : \mathbb{R}^k \rightarrow L^2(\Omega, \mu)$ be a bounded linear map. Put $s_N = 2 + \lceil \log_2(N+1) \rceil$. Then, for every $x \in \mathbb{R}^k$,*

$$\int \max_{0 \leq j < N} |\mathcal{T}\Pi_j x|^2 d\mu \leq s_N^2 \|\mathcal{T}\|^2 \|x\|^2. \quad (6.1)$$

Proof. The differences $P_j = \Pi_{j-1} - \Pi_j$ are mutually orthogonal projectors. Pad the sequence with zero projectors to a power-of-two length. Each tail sum $\Pi_j = \sum_{h>j} P_h$ can be written as a sum of at most s_N operators $P_I = \sum_{h \in I} P_h$, where the index sets I are disjoint dyadic intervals. Pointwise Cauchy–Schwarz gives

$$\max_j |\mathcal{T}\Pi_j x|^2 \leq s_N \sum_{I \text{ dyadic}} |\mathcal{T}P_I x|^2.$$

At each dyadic level the interval projectors have orthogonal ranges, so

$$\sum_{I \text{ at that level}} \|\mathcal{T}P_I x\|_{L^2}^2 \leq \|\mathcal{T}\|^2 \sum_I \|P_I x\|^2 = \|\mathcal{T}\|^2 \|x\|^2.$$

There are at most s_N levels. Integrating proves (6.1). Orthogonality is needed in coefficient space; the functions $\mathcal{T}P_j x$ need not be orthogonal. $\square$

The classical Rademacher–Menshov maximal inequality controls maximal partial sums of orthogonal functions; see [7, Proposition 2.1]. The proof above adapts the usual dyadic argument to a bounded operator on coefficient space and makes the dependence on N explicit.

Lemma 6.2. *For every $x \in \mathbb{R}^k$, the maps in (5.10) satisfy*

$$\sum_i \mathcal{E}_{S_i}(F_i x) \leq s_N^2 \lambda_k \|x\|^2. \quad (6.2)$$

Proof. Fix an orientation of each edge and apply Lemma 6.1 to the edge-gradient map

$$(\mathcal{T}x)(\{u, v\}) = \sqrt{w_{uv}} \langle \mathbf{f}_u - \mathbf{f}_v, x \rangle,$$

with counting measure on edges. Equation (2.1) gives $\|\mathcal{T}\|^2 \leq \lambda_k$. Every edge lies inside at most one selected set. Its contribution to the left side of (6.2) is therefore at most the corresponding pointwise maximum in (6.1). $\square$

The final lemma converts the bounds on the Gram matrices and internal energies into exactly k sets. We use the dimension argument of [4], with each local eigenvalue count truncated at two, and apply it to the degree-preserving local operators instead of Dirichlet operators.

Lemma 6.3 (Counting local eigenvalues). *Let $\tau \geq 0$, and let $F_i : \mathbb{R}^k \rightarrow L^2(d)$ be linear maps supported on pairwise disjoint nonempty sets S_i . Suppose each S_i is τ -stable and that, for some $0 \leq \delta < \gamma$ and $\Lambda \geq 0$,*

$$\sum_i F_i^* F_i \succeq \gamma I_k, \quad \text{tail}_1(F_i^* F_i) \leq \delta, \quad \sum_i \mathcal{E}_{S_i}(F_i x) \leq \Lambda \|x\|^2 \quad (x \in \mathbb{R}^k).$$

Then

$$\varphi_k(G) \leq \tau + 4\sqrt{\Lambda/(\gamma - \delta)}. \quad (6.3)$$

Proof. First assume $\Lambda > 0$ and put $\sigma = 2\Lambda/(\gamma - \delta)$. Let c_i be the number of local eigenvalues $\lambda_j(S_i)$ at most σ , truncated at two. Since $\lambda_1(S_i) = 0$, every c_i is at least one. We claim that $\sum_i c_i \geq k$.

Suppose instead that the sum is less than k , and let p be the number of sets. Since each c_i is either one or two, $p < k$ and fewer than $k - p$ indices satisfy $c_i = 2$. For each i with $c_i = 1$, impose on x the linear constraint that $F_i x$ be orthogonal, in $L^2(d|_{S_i})$, to a unit vector spanning the one-dimensional local spectral subspace for eigenvalues at most σ . For each i with $c_i = 2$, choose a unit eigenvector u_i corresponding to the largest eigenvalue of $F_i^* F_i$, and require that x be orthogonal to u_i . These are p homogeneous linear constraints. Their common kernel contains $k - p$ orthonormal vectors $x_1, \dots, x_{k-p}$.

For $c_i = 1$, the spectral theorem gives $\|F_i x_a\|_d^2 \leq \mathcal{E}_{S_i}(F_i x_a)/\sigma$. For $c_i = 2$, the matrix $\sum_a x_a x_a^T$ is the orthogonal projector onto the span of the vectors x_a . Each x_a is orthogonal to u_i , so $0 \leq \sum_a x_a x_a^T \preceq I_k - u_i u_i^T$. Consequently,

$$\sum_a \|F_i x_a\|_d^2 = \text{tr}\left(F_i^* F_i \sum_a x_a x_a^T\right) \leq \text{tr}(F_i^* F_i (I_k - u_i u_i^T)) = \text{tail}_1(F_i^* F_i) \leq \delta.$$

Thus δ bounds the sum of the squared $L^2(d)$ -norms of all $k - p$ functions $F_i x_a$. Summing over i and a and using the Gram matrix and energy hypotheses gives

$$\gamma(k - p) \leq \frac{\Lambda(k - p)}{\sigma} + \delta(k - p) = \frac{\gamma + \delta}{2}(k - p) < \gamma(k - p),$$

a contradiction.

For a set contributing one to the count, use S_i itself; its conductance is at most τ . For one contributing two, $\lambda_2(S_i) \leq \sigma$. If the local graph is disconnected, use two of its connected components, each of local conductance zero. Otherwise, a second eigenfunction f is orthogonal to the constant function and has two nonzero sign parts. The pointwise contraction inequality gives

$$\mathcal{E}_{S_i}(f_+) \leq \langle f_+, \Delta_{S_i} f \rangle_d = \lambda_2(S_i) \|f_+\|_d^2.$$

Applying the same estimate to $-f$ gives the bound for f_- . Cheeger rounding in the local graph gives two disjoint nonempty sets of local conductance at most $\sqrt{2\sigma}$. Equation (3.2) bounds their original conductance by $\tau + 2\sqrt{2\sigma}$, which is (6.3). Since $\sum_i c_i \geq k$, retain any k of the resulting disjoint sets. If $\Lambda = 0$, apply the positive- Λ case with any $\Lambda' > 0$, for which the energy hypothesis still holds. Let Λ' tend to zero and pass to the limit over the finite family of possible sets. $\square$

Proof of Theorem 1.1. For $k = 1$, take V . If $\lambda_k = 0$, the graph has at least k connected components, each of positive volume; use any k of them. Otherwise apply the construction of Section 5. Lemmas 5.2 and 6.2 verify the hypotheses of Lemma 6.3 with the threshold τ from (5.1), $\gamma = \varepsilon$, $\delta = \varepsilon/4$, and $\Lambda = s_N^2 \lambda_k$. Thus

$$\varphi_k(G) \leq C_0 \sqrt{\lambda_k \log(k + 1)} + \frac{4s_N \sqrt{\lambda_k}}{\sqrt{\varepsilon - \delta}}.$$

By (5.9), $N = O(\log(k + 1))$. Consequently

$$s_N = O(1 + \log \log(k + 2)) = O(\sqrt{\log(k + 1)}),$$

which proves (1.3). $\square$

7 Sharpness of the logarithmic factor

We recall the noisy-hypercube example from [3, Section 4.4] to check that the lower bound uses the same spectral index.

Proposition 7.1. *For every integer $k \geq 3$, there is a weighted graph G such that*

$$\frac{\varphi_k(G)}{\sqrt{\lambda_k(G)}} \geq \sqrt{\frac{\log k}{8 \log 2}}.$$

Proof. Fix $k \geq 3$. Let G be the graph on $\{0, 1\}^k$ with weights $w_{xy} = \zeta^{|x-y|}$, including loops, where $|x-y|$ is Hamming distance and $\zeta = (\log 2)/\log k$. The common degree is $(1+\zeta)^k$. The transition operator is the Boolean noise operator T_ρ with correlation $\rho = (1-\zeta)/(1+\zeta)$. Its eigenvalue on each Walsh character of degree j is ρ^j . Thus the first nonzero Laplacian eigenvalue has multiplicity k and

$$\lambda_k(G) = \frac{2\zeta}{1+\zeta}.$$

Let μ be the uniform probability measure. For every nonempty $S \subseteq \{0, 1\}^k$, the hypercontractive estimate used in [3, Theorem 4.12] gives

$$\frac{\langle \mathbf{1}_S, T_\rho \mathbf{1}_S \rangle_\mu}{\mu(S)} \leq \mu(S)^{(1-\rho)/(1+\rho)} = \mu(S)^\zeta.$$

Equivalently, $\varphi(S) \geq 1 - \mu(S)^\zeta$. Indeed, hypercontractivity gives

$$\langle \mathbf{1}_S, T_\rho \mathbf{1}_S \rangle_\mu = \|T_{\sqrt{\rho}} \mathbf{1}_S\|_{L^2(\mu)}^2 \leq \|\mathbf{1}_S\|_{L^{1+\rho}(\mu)}^2.$$

Among k disjoint nonempty sets, one has measure at most $1/k$, and therefore conductance at least $1 - k^{-\zeta} = 1/2$. It follows that

$$\frac{\varphi_k(G)}{\sqrt{\lambda_k(G)}} \geq \frac{1}{2} \sqrt{\frac{1+\zeta}{2\zeta}} \geq \sqrt{\frac{\log k}{8 \log 2}}.$$

□

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